

100% Excellent!

Name: _____

These questions assess your understanding of the material from Chapter(s) 23, 24 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. A portable x-ray unit has a step-up transformer, the 100 V input of which is transformed to the 2.5×10^5 V output needed by the x-ray tube. The primary has 56 loops and draws a current of 1.4 A when in use. Find the current output of the secondary.

ANSWER: 5.6×10^{-9} A ✓

$$E_P = 100 \text{ V}$$

$$E_S = 2.5 \times 10^5 \text{ V}$$

$$I_P = 1.4 \text{ A}$$

$$I_S = ?$$

$$\frac{V_S}{V_P} = \frac{I_P}{I_S}$$

$$\frac{2.5 \times 10^5}{100} = \frac{1.4}{I_S}$$

$$I_S = 5.6 \times 10^{-9} \text{ A}$$

2. A generator coil is rotated 0.10 revolutions in 32. ms. The 110-turn circular coil has a 2.0-cm radius and is in a uniform 1.7-T magnetic field. What is the average emf that is induced?

ANSWER: 7.34 V ✓

$$\theta = 0.10 \text{ rev} \left| \frac{2\pi \text{ rad}}{1 \text{ rev}} \right. = 0.63 \text{ rad}$$

$$t = 32 \times 10^{-3} \text{ s}$$

$$N = 110 \text{ turns}$$

$$r = 2.0 \times 10^{-2} \text{ m}$$

$$B = 1.7 \text{ T}$$

$$\Phi = BA \Delta \theta$$

$$= (1.7)(1.26 \times 10^{-3})(\cos(90) - \cos(0))$$

$$= -2.136 \times 10^{-3} \text{ Wb}$$

$$A = \pi r^2$$

$$= (\pi)(2.0 \times 10^{-2})^2 = 1.26 \times 10^{-3} \text{ m}^2$$

$$\text{emf} = \frac{-N \Delta \Phi}{\Delta t}$$

3. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of 1.1×10^3 V/m?

ANSWER: 3.67×10^{-6} T ✓

$$c = \frac{E}{B}$$

$$B = \frac{E}{c}$$

$$= \frac{1.1 \times 10^3}{3.0 \times 10^8}$$

$$= 3.67 \times 10^{-6} \text{ T}$$

+3

✓

4. How much energy is stored in a 0.28-mH inductor when a 11.-A current flows through it?

ANSWER: 0.01694 J

$$E_{ind} = \frac{1}{2} L I^2$$

$$= \frac{1}{2} (0.28 \times 10^{-3}) (11)^2$$

$$= 0.01694 \text{ J}$$

✓

5. What is the characteristic time constant for a 13.-mH inductor in series with a 5.6-Ω resistor?

ANSWER: $2.32 \times 10^{-3} \text{ s}$

$$\tau = \frac{L}{R}$$

$$= \frac{13 \times 10^{-3}}{5.6}$$

$$= 2.32 \times 10^{-3} \text{ s}$$

✓

6. A battery charger meant for a series connection of several nickel-cadmium batteries needs to have a 17.-V output to charge the batteries. It uses a step-down transformer with a 220-loop primary and a 110-V input. If the charging current is 35. A, what is the input current?

ANSWER: 5.41 A

$$V_s = 17 \text{ V}$$

$$V_p = 110 \text{ V}$$

$$I_s = 35 \text{ A}$$

$$\frac{V_s}{V_p} = \frac{I_p}{I_s}$$

$$\frac{17}{110} = \frac{I_p}{35}$$

$$I_p = 5.41 \text{ A}$$

7. Suppose a 150-turn coil lies in the plane of the page in a uniform magnetic field that is directed into the page. The coil originally has an area of 1.1 m^2 . It is then stretched for 0.38 s until it has no area. What is the magnitude of the induced emf if the uniform magnetic field has a strength of 0.11 T ?

ANSWER: 47.76 V ✓

$$N = 150 \text{ turn}$$

$$A_i = 1.1 \text{ m}^2 \quad A_f = 0 \text{ m}^2$$

$$t = 0.38 \text{ s}$$

$$B = 0.11 \text{ T}$$

$$\begin{aligned} \Phi &= BA \cos \theta \\ &= (0.11)(0 - 1.1) \\ &= -0.121 \text{ Wb} \end{aligned}$$

$$\begin{aligned} \text{emf} &= -\frac{N \Delta \Phi}{\Delta t} \\ &= \frac{-(150)(-0.121)}{0.38} \\ &= 47.76 \text{ V} \end{aligned}$$

8. A generator coil is rotated 0.46 revolutions in 39 ms . The 400 -turn circular coil has a 3.8-cm radius and is in a uniform 1.4-T magnetic field. What is the maximum emf that is induced?

ANSWER: 188.27 V ✓

$$\theta = 0.46 \text{ rev} \left| \frac{2\pi \text{ rad}}{1 \text{ rev}} \right. = 2.89 \text{ rad}$$

$$t = 39 \times 10^{-3} \text{ s}$$

$$N = 400 \text{ turn}$$

$$r = 3.8 \times 10^{-2} \text{ m}$$

$$B = 1.4 \text{ T}$$

$$\begin{aligned} \text{emf} &= NAB\omega \sin(\omega t) \\ &= (400)(\pi)(3.8 \times 10^{-2})^2(1.4)\left(\frac{2.89}{39 \times 10^{-3}}\right) \\ &= 188.27 \text{ V} \end{aligned}$$

9. Calculate the motional emf induced along a 7.6-km -long conducting object moving at an orbital speed of $5.4 \frac{\text{km}}{\text{s}}$ perpendicular to Earth's $29\text{-}\mu\text{T}$ magnetic field.

ANSWER: 1190.16 V ✓

$$l = 7.6 \times 10^3 \text{ m}$$

$$v = 5.4 \frac{\text{km}}{\text{s}} \left| \frac{1000 \text{ m}}{1 \text{ km}} \right. = 5400 \text{ m/s}$$

$$B = 29 \times 10^{-6} \text{ T}$$

$$\begin{aligned} \mathcal{E} &= Blv \\ &= (29 \times 10^{-6})(7.6 \times 10^3)(5400) \\ &= 1190.16 \text{ V} \end{aligned}$$

+3

10. On its highest power setting, a 1.3×10^3 -W microwave oven projects microwaves onto a 0.43-m-by-0.43-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.

ANSWER: 2301.4 V/m

$$I_{\text{ave}} = \frac{P}{A}$$

$$= \frac{1.3 \times 10^3}{(0.43 \times 0.43)}$$

$$= 7030.83 \text{ A}$$

$$I_{\text{ave}} = \frac{c \epsilon_0 E_0^2}{2}$$

$$E_0 = 2301.37$$

11. Suppose a 1.5-H inductor is in series with a 1.4-Ω resistor and a current of 11 A flows through them. Find the current that flows through the inductor and resistor 1.8 s after the battery is disconnected.

ANSWER: 2.05 A

$$I(t) = I_0 e^{-t/\tau}$$

$$\tau = \frac{L}{R}$$

$$= \frac{1.5}{1.4}$$

$$= 1.07 \text{ s}$$

$$I = (11) e^{-1.8/1.07}$$

$$= 2.05 \text{ A}$$

12. Calculate the self-inductance of a 0.27 m-long, 0.024 m-diameter solenoid that has 390 coils.

ANSWER: $3.20 \times 10^{-4} \text{ H}$

$$L = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{(4\pi \times 10^{-7})(390)^2(4.52 \times 10^{-4})}{0.27}$$

$$= 3.20 \times 10^{-4} \text{ H}$$

$$\frac{0.024}{2} = 0.012 \text{ m}$$

$$A = \pi r^2$$

$$1. I_s = 5.60 \times 10^{-4} \text{ A} \because \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(Chapter 23, Section (7) Transformers - cple_ch23_ex5)

$$2. \varepsilon = 7.34 \text{ V} \because \varepsilon = -N \frac{\Delta\Phi}{\Delta t}$$

(Chapter 23, Section (5) Electric Generators - cple_ch23_ex3)

$$3. B_{max} = 3.67 \times 10^{-6} \text{ T} \because c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f\lambda$$

(Chapter 24, Section (2) Production of Electromagnetic Waves - cple_ch24_ex1)

$$4. E_{ind} = 0.0169 \text{ J} \because E_{ind} = \frac{1}{2} LI^2$$

(Chapter 23, Section (9) Inductance - cple_ch23_ex8)

$$5. \tau = 2.32 \times 10^{-3} \text{ s} \because \tau = \frac{L}{R}$$

(Chapter 23, Section (10) RL Circuits - cple_ch23_ex9)

$$6. I_p = 5.41 \text{ A} \because \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(Chapter 23, Section (7) Transformers - cple_ch23_ex6)

$$7. \text{emf} = 47.8 \text{ V} \because \text{emf} = -N \frac{\Delta\Phi}{\Delta t}$$

(Chapter 23, Section (2) Faraday's Law of Induction: Lenz's Law - cple_ch23_p8)

$$8. \varepsilon = 188. \text{ V} \because \varepsilon = NAB\omega \sin(\omega t), \sin(\omega t) = 1$$

(Chapter 23, Section (5) Electric Generators - cple_ch23_ex4)

$$9. \varepsilon = 1.19 \times 10^3 \text{ V} \because \varepsilon = B\ell v$$

(Chapter 23, Section (3) Motional EMF - cple_ch23_ex2)

$$10. E_0 = 2.30 \times 10^3 \frac{\text{V}}{\text{m}} \because I_{ave} = \frac{c\varepsilon_0 E_0^2}{2} = \frac{cB_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$$

(Chapter 24, Section (4) Energy in Electromagnetic Waves - cple_ch24_ex4)

$$11. I = 2.05 \text{ A} \because I(t) = I_0 e^{-t/\tau}, \tau = \frac{L}{R}$$

(Chapter 23, Section (10) RL Circuits - cple_ch23_ex9b)

$$12. L = 3.20 \times 10^{-4} \text{ H} \because L = N \frac{\Delta\Phi}{\Delta t} = \frac{\mu_0 N^2 A}{\ell}$$

(Chapter 23, Section (9) Inductance - cple_ch23_ex7)